

- 7/7/26 Task-2.
- 1) Top-product based companies (DSA + heavy)
- Google, Amazon, Microsoft, Apple, Netflix.
  - Adobe, FinTech, SaaS etc.

Major Indian service-based companies (use DSA in tests)

- TCS, Infosys, Wipro, Cognizant, HCL.
- Tech Mahindra, Capgemini, Accenture, DXC Technology, CGI.

Popular Indian product/startup companies using DSA rounds.

- Flipkart, Ola, Swiggy, Zomato, Paytm, Razor pay.
- Zoho, Inmob, Redbus, Make my trip.

2) Gate syllabus for DSA

=> programming and Data structures.

- Programming basics: programming inc, variables, control flow, functions, recursion.
- Linear data structures: Arrays, strings, stacks, queues, linked lists.
- Trees: General trees, binary, binary search trees, basic operators.
- Graphs: Representation, basic graph theory concept.
- Hashing: Hash tables, hash functions, collision handling.

## Algorithms:

- Searching: linear & binary
- Sorting - basic: Bubble, selection & insertion
- Sorting - advanced: merge, quick, heap-sort
- Complexity: Asymptotic worst-case time and space complexity, Big-O,  $\Omega$ ,  $\Theta$
- Algorithm design techniques: Greedy, dynamic programming, divide-and-conquer
- Graph algorithms.

3)  $\rightarrow$  Message to a friend

### Inputs:

- Sender's name/number
- Receiver's name/number
- Message content
- Timestamp

### Outputs:

- Delivered message
- Delivery Status
- Timestamp of delivery

2) Bus ticket

Inputs

- present station
- destination station
- Date of travel
- no of passengers
- Bus Type

Outputs

- Ticket number
- Seat number
- fare amount
- Departure time
- Destination point
- Total amount to paid

3) Electricity Bill

Inputs

- consumer ID
- previous meter reading
- current meter reading
- no of units consumed
- Rate per unit
- meter rent
- due date

Outputs

- units consumed
- Total Bill
- Due date for payment
- Late payment penalty

#### 4) The Real-life Examples of DSA

• Queue - Students are waiting at canteen counter

Stack - Books piled on your desk

Array - houses & shops in a row

Tree - hierarchy in my college

Graph - the road network itself! Roads connecting hostel, college, bus stop

Hash-table - my collage Id card (or) roll number system.

2D array / matrix - college timetable.